Getting philosophical

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Causal mapping and philosophy

Some thoughts...

Context and the transitivity trap

> From (Powell et al., 2024)

Transitivity is perhaps the single most important challenge for causal mapping. Consider the following example. If source P [pig farmer] states 'I received cash grant compensation for pig diseases [G], so I had more cash [C]', and source W [wheat farmer] states 'I had more cash [C], so I bought more seeds [S]', can we then deduce that pig diseases lead to more cash which leads to more seed (G à C àS), and therefore G à S (there is evidence for an indirect effect of G on S, i.e. that cash grants for pig diseases lead to people buying more seeds)?

The answer is of course that we cannot because the first part only makes sense for pig farmers, and the second part only makes sense for wheat farmers.

In general, from G --> C (in context P) and C --> S (in context W), we can only conclude that G --> S in the intersection of the contexts P and W. Correctly making inferences about indirect effects is the key benefit but also the key challenge for any approach which uses causal diagrams or maps, including quantitative approaches (Bollen, 1987).

References

Powell, Copestake, & Remnant (2024). *Causal Mapping for Evaluators*. https://doi.org/10.1177/13563890231196601.

Context, mechanisms and triggers 2

If termites cause a tree to fall in a forest where no-one can hear it, was it a causal event?

Maybe not.

The difference is the need for an explanation. That's the trigger. The trigger says why here and now if forest fires are happening all the time then we lose the sense of the trigger and the explanation could involve anything Oxygen or gravity or whatever. We don't have a question to answer so we don't know what factors to mention right the way out to the orbits of Pluto and beyond

Context, mechanisms and triggers

> What delineates a mechanism and its contents, the circle in the CMO diagram, is not so much facts about the world but the things which I happen to need to call upon to make a causal explanation

I found the concept of a trigger in realist evaluation totally baffling because RE is supposed to be somehow scientific, yet most forms of scientific explanation don't involve actual triggers.

But then I realised, triggers (and mechanisms) are best understood from an epistemic perspective.

"This mechanism gets triggered here" can be parsed as: there is a need for an explanation here: "I am invoking this mechanism to explain something that needs explaining".

We don't invoke oxygen to explain the forest fire, although both oxygen and the spark are (let's say) necessary.

What does a causal coding mean

The way we do causal mapping, a coded causal claim does not mean:

- "X occurred and then Y occurred"
- "Y frequently occurs after X"
- "X has the causal power to affect Y"

It means something like

- "X affected Y via its causal power to do so"

There is no hidden vocabulary

Our approach is minimalist -- we code only bare causation

It follows from this that we are sceptical about the idea of more sophisticated mid-range theory with blockers, enablers etc.

QCA is disappointing because it is frequentist about causation

TODO

QCA is disappointing because it thinks the world is a grid

TODO

[1c A minimalist approach to coding does not code absences]

Counterfactuals are part of the meaning of causation but are not necessarily part of how we know about it

Just about everything is complex

TODO

> Putting a man on the moon was merely a complicated task, not a complex one?

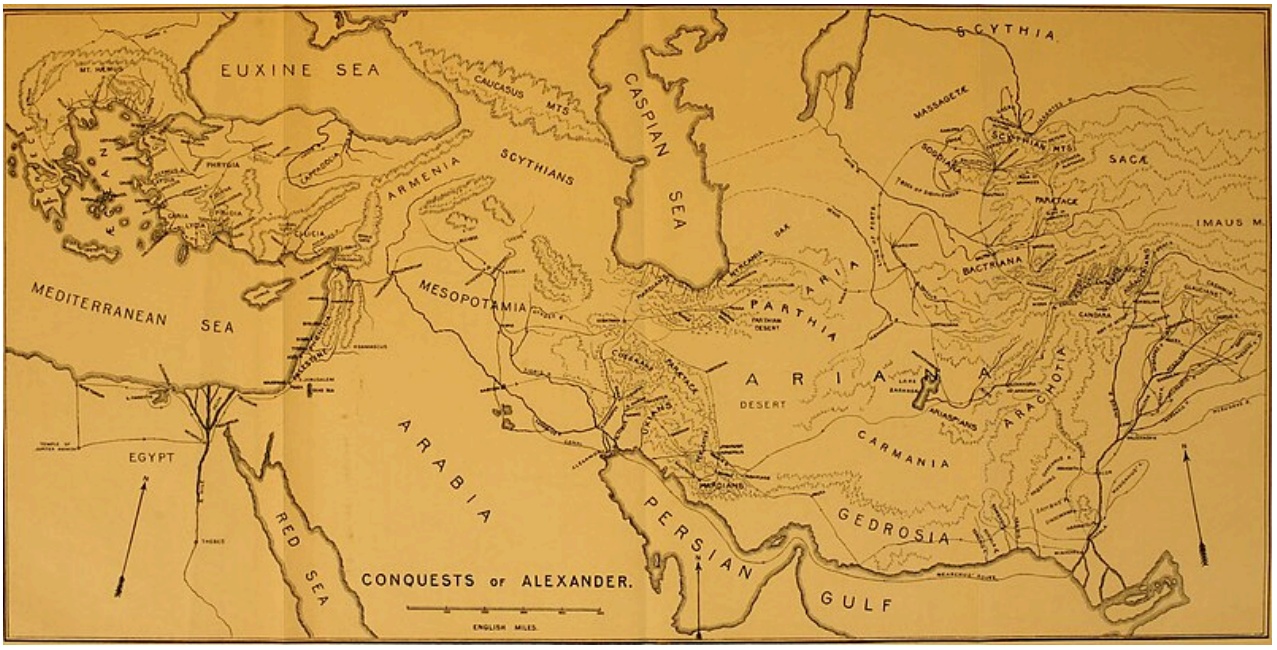
“Putting a man on the moon” is really often given as an example of a merely complicated but not complex task - Glouberman and Zimmerman (2002), cited in Rogers (2008).

But while that task was certainly very complicated, it was often complex too. There were plenty of conflicting sub-goals, arguments about how to solve a particular problem, interdependencies and conflicts between means, and between ends… You can *see* it as merely complicated, in order to make a point, and maybe that’s OK. But if you’re an actual space scientist you’d probably disagree.

There has always been complexity

Irene Ng speaks for many who write about "complex systems" when she says: "What has happened in the last 50 years is that we've been trying to use deterministic tools to achieve emergent outcomes, essentially because those are the only tools we have learnt (systems thinkers are still a minority unfortunately). We treat complex systems like complicated systems. We try to design, specify, impose, dictate when we should be designing, enabling, intervening, stablising."

Is there any historical truth in this at all? Did, say, a midwife 50 years ago only know how to impose and dictate rather than intervene and stablise? Was, say, managing the Mongol Empire, or Alexander's conquests, a merely complicated, not complex task?



An often-used example of a complex task is bringing up a child (and I'd agree, loosely). Well, did we have no children to bring up until our frightfully modern era?

Perhaps Irene Ng is writing about *our writing about* management, not how it is or was actually done. But there are ancient books like "The Art of War" about how to lead, and manage, and reach goals. Were they all merely guides to snapping together simple solutions? Of course not.